

Critical: The simulated 0-5 V sensor signal goes to **J2 Pin 3 (WIND_RAW)**. Do not apply it to J2 Pin 1. J2 Pin 1 is the fixed +12 V sensor-power rail. Never apply 12 V to WIND_RAW, WIND_ADC, or the ESP32.

1. Connector identification

Connector	Pin	Function	Allowed voltage
J1	1	PCB/sensor +12 V input	Nominal 12 V
J1	2	Ground	0 V
J2	1	+12 V output to real sensor	Nominal 12 V
J2	2	Ground	0 V
J2	3	WIND_RAW / simulated sensor signal	0-5.00 V maximum
J2	4	Not connected	Do not use

Confirm physical pin numbering from the PCB silkscreen or design before connecting power.

2. Initial wiring - ESP32 disconnected

Turn off all supply outputs before making or changing connections.

12 V supply + -----> J1 Pin 1 (+12 V)

12 V supply - -----> J1 Pin 2 (GND)

Variable supply + -----> J2 Pin 3 (WIND_RAW)

Variable supply - -----> J2 Pin 2 (GND)

J2 Pin 1 = +12 V sensor rail; leave unused during signal simulation

J2 Pin 4 = NC; leave disconnected

The 20 kOhm/10 kOhm signal divider is passive, so 12 V is not required merely to test the divider. The 12 V supply may be connected when checking the complete PCB, but it does not generate the simulated signal.

3. Configure the variable supply

1. Keep the output switched **OFF**.
 2. Set voltage to **0.00 V**.
 3. Set the current limit to approximately **10-20 mA**.
 4. Verify positive is connected to **J2 Pin 3**.
 5. Verify negative is connected to **J2 Pin 2/GND**.
 6. Do not set the simulated signal above **5.00 V**.
- The divider normally draws only about 0.17 mA at a 5 V input.

4. Check WIND_RAW and WIND_ADC

1. Set the multimeter to DC volts.
2. Connect the black probe to J2 Pin 2/GND.
3. Connect the red probe to J2 Pin 3/WIND_RAW.
4. Switch on the variable output at 0.00 V, then set 0.50 V.
5. Confirm WIND_RAW is approximately 0.50 V.
6. Move only the red probe to the WIND_ADC test point or D1/GPIO3 header pad.
7. Confirm WIND_ADC is approximately WIND_RAW divided by three.

Pass condition: $\text{WIND_ADC} = \text{WIND_RAW} / 3$, allowing for normal resistor and meter tolerance.

5. Expected test readings

J2 Pin 3 / WIND_RAW	Expected WIND_ADC	Simulated wind	Simulated wind
0.00 V	0.000 V	0 m/s	0.0 km/h
0.50 V	0.167 V	3 m/s	10.8 km/h
1.00 V	0.333 V	6 m/s	21.6 km/h
2.00 V	0.667 V	12 m/s	43.2 km/h
3.00 V	1.000 V	18 m/s	64.8 km/h
4.00 V	1.333 V	24 m/s	86.4 km/h
5.00 V	1.667 V	30 m/s	108.0 km/h

6. Connect and power the ESP32

Proceed only after the divider readings pass.

1. Switch off all supply outputs and return the variable supply to 0.00 V.
2. Connect PCB WIND_ADC to XIAO **D1/GPIO3**.
3. Connect XIAO GND to PCB GND.
4. Power the XIAO from USB, or from a properly regulated supply connected to its designated 5 V input.
5. Avoid simultaneously powering it from USB and an external 5 V source unless safe isolation is confirmed.

12 V supply negative -----+

Variable supply negative -----+----- PCB GND ---- XIAO GND

ESP32 supply negative -----+

PCB WIND_ADC ----- XIAO D1 / GPIO3

Regulated 5 V or USB ----- XIAO power

PCB +12 V ----- NEVER connect to XIAO

7. Firmware verification

Open the serial monitor at **115200 baud**. Increase the simulated signal gradually.

Example: 2.00 V at J2 Pin 3

WIND_ADC=0.667 V | SENSOR=2.000 V | WIND=12.00 m/s | 43.2 km/h

Example: 5.00 V at J2 Pin 3

WIND_ADC=1.667 V | SENSOR=5.000 V | WIND=30.00 m/s | 108.0 km/h

Small differences between the multimeter and ESP32 readings are normal.

8. Stop conditions and safety

- Stop if WIND_ADC is not approximately one-third of WIND_RAW.
- Stop if WIND_ADC exceeds approximately 1.7 V with a maximum 5 V input.
- Stop if a supply enters current limiting or any component becomes warm.
- Never use current/A mode with the meter connected directly across a supply.
- Perform resistance and continuity tests only while all power is disconnected.
- Never apply 12 V to J2 Pin 3, WIND_ADC, D1/GPIO3, or the XIAO 5 V pin.

9. Test record

Set WIND_RAW	Measured WIND_RAW	Measured WIND_ADC	Serial result	Pass
0.00 V				
0.50 V				
1.00 V				
2.00 V				
3.00 V				
4.00 V				
5.00 V				

SPL Wind Speed Logger - bench procedure based on $R1 = 20\text{ k}\Omega$, $R2 = 10\text{ k}\Omega$ and $C1 = 100\text{ nF}$.